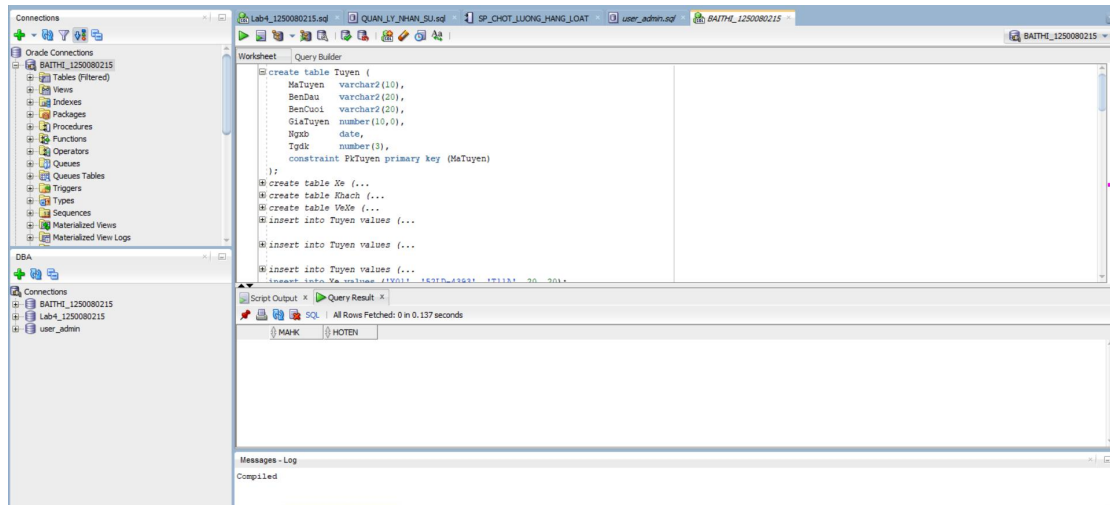


Tên : Lê Quốc Trung  
Lớp :12CNTT4  
MSSV : 1250080215



```
create table Tuyen (  
    MaTuyen varchar2(10),  
    BenDau varchar2(20),  
    BenCuoi varchar2(20),  
    GiaTuyen number(10,0),  
    Ngay date,  
    Tgdk number(3),  
    constraint PkTuyen primary key (MaTuyen)  
);  
create table Xe (  
    MaXe varchar2(10),  
    BienKs varchar2(20),  
    MaTuyen varchar2(10),  
    SoGheT1 number(3),  
    SoGheT2 number(3),  
    constraint PkXe primary key (MaXe),  
    constraint FkXeTuyen foreign key (MaTuyen)  
        references Tuyen(MaTuyen)  
);  
create table Khach (  
    MaHk varchar2(10),  
    HoTen varchar2(50),  
    GioiTinh varchar2(5),  
    Cmnd varchar2(15),  
    constraint PkKhach primary key (MaHk)  
);  
create table VeXe (  
    MaTuyen varchar2(10),  
    MaHk varchar2(10),  
    NgMua date,  
    GiaVe number(10,0),  
    constraint PkVeXe primary key (MaTuyen, MaHk, NgMua),  
    constraint FkVeXeTuyen foreign key (MaTuyen)  
        references Tuyen(MaTuyen),  
    constraint FkVeXeKhach foreign key (MaHk)  
        references Khach(MaHk)  
);  
insert into Tuyen values (  
    'T11A', 'SG', 'DL', 210000,  
    to_date('26/12/2016','dd/mm/yyyy'), 6  
);  
insert into Tuyen values (  

```

```

'T32D', 'PT', 'SG', 120000,
to_date('30/12/2016','dd/mm/yyyy'), 4
);

insert into Tuyen values (
'T06F', 'NT', 'DNG', 225000,
to_date('02/01/2017','dd/mm/yyyy'), 7
);
insert into Xe values ('X01', '52LD-4393', 'T11A', 20, 20);
insert into Xe values ('X02', '59LD-7247', 'T32D', 36, 36);
insert into Xe values ('X03', '55LD-6850', 'T06F', 15, 15);
insert into Khach values ('KH01', 'Lâm Văn Bền', 'Nam', '655615896');
insert into Khach values ('KH02', 'Đương Thị Lục', 'Nữ', '275648642');
insert into Khach values ('KH03', 'Hoàng Thanh Tùng', 'Nam', '456889143');
insert into VeXe values (
'T11A', 'KH01',
to_date('20/12/2016','dd/mm/yyyy'), 210000
);

insert into VeXe values (
'T32D', 'KH02',
to_date('25/12/2016','dd/mm/yyyy'), 144000
);

insert into VeXe values (
'T06F', 'KH03',
to_date('30/12/2016','dd/mm/yyyy'), 270000
);

```

```

---3---
alter table Tuyen
add constraint CkTuyenGia
check (Tgdk <= 5 or GiaTuyen > 200000);

```

```

---4---
create or replace trigger TrgTangGia
before insert or update on Tuyen
for each row
begin
if :new.Ngxb between
to_date('29/12/2016','dd/mm/yyyy')
and
to_date('05/01/2017','dd/mm/yyyy')
then
:new.GiaTuyen := :new.GiaTuyen * 1.2;
end if;
end;

```

```

---5---
select *
from VeXe
where extract(month from NgMua) = 12
order by GiaVe desc;

```

|   | ⚡ MATUYEN | ⚡ MAHK | ⚡ NGMUA     | ⚡ GIAVE |
|---|-----------|--------|-------------|---------|
| 1 | T06F      | KH03   | 30-DEC-2016 | 270000  |
| 2 | T11A      | KH01   | 20-DEC-2016 | 210000  |
| 3 | T32D      | KH02   | 25-DEC-2016 | 144000  |

```

---6---
select MaTuyen
from VeXe
where extract(year from NgMua) = 2016
group by MaTuyen
having count(*) = (

```

```

select min(count(*))
from VeXe
where extract(year from NgMua) = 2016
group by MaTuyen
);

```

|   | MATUYEN |
|---|---------|
| 1 | T06F    |
| 2 | T11A    |
| 3 | T32D    |

---7---

```

select V.MaTuyen
from VeXe V
join Khach K on V.MaHk = K.MaHk
group by V.MaTuyen
having
    sum(case when K.GioiTinh = 'Nam' then 1 else 0 end) > 0
and
    sum(case when K.GioiTinh = 'Nữ' then 1 else 0 end) > 0;

```

|  | MATUYEN |
|--|---------|
|--|---------|

---8---

```

select K.MaHk, K.HoTen
from Khach K
where K.GioiTinh = 'Nữ'
and not exists (
    select T.MaTuyen
    from Tuyen T
    where not exists (
        select 1
        from VeXe V
        where V.MaTuyen = T.MaTuyen
        and V.MaHk = K.MaHk
    )
);

```

| MAHK | HOTEN |
|------|-------|
|------|-------|